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Practice Examination, Semester 2, 2025

Evolutionary Computation COMP SCI 3316/7316
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Course Coordinator: Mr Anthropic Claude

Total Duration: 180 mins

Questions	Time	Marks
Answer all 25 questions	180 mins	150 marks
		150 Total

Instructions for Candidates

- This is a Modified Closed-Book examination (refer to Materials)
- Write all answers within the space under the question on the exam paper
- If you need more space, use the extra blank pages at the end of the exam
- Write your desk number on each page
- Examination material must not be removed from the examination room

Materials permitted:

- Paper dictionary - general or translation
- One double sided A4 pages of notes

DO NOT COMMENCE WRITING UNTIL INSTRUCTED TO DO SO

STOP WRITING IMMEDIATELY WHEN INSTRUCTED

Section 1: Local Search Algorithms [25 marks]

Question 1: Hill Climbing vs Simulated Annealing [15 marks]

Consider the following fitness landscape where we want to **maximize** the fitness function $f(x)$:

Position x	0	1	2	3	4	5	6
Fitness $f(x)$	10	15	12	20	18	25	22

Assume the neighborhood structure is: $N(x) = \{x-1, x+1\}$ (with boundary: $N(0) = \{1\}$ and $N(6) = \{5\}$).

- (a) Starting from position $x = 1$, perform **Hill Climbing (steepest ascent)** until convergence. Show each step, the neighbors evaluated, and explain why the algorithm stops.

[4 marks]

(b) Now starting from the same position $x = 1$, perform **Simulated Annealing** for 5 iterations using:

- Initial temperature $T_0 = 10$
- Cooling schedule: $T_{t+1} = 0.8 \cdot T_t$
- Acceptance probability: $P(\text{accept worse}) = e^{-\Delta f / T}$ where $\Delta f = f(x_{\text{current}}) - f(x_{\text{new}})$
- Random neighbor selection sequence: $[2, 0, 2, 2, 1]$ (which neighbor to try)
- Random acceptance values: $[0.3, 0.7, 0.2, 0.9, 0.1]$ (for deciding acceptance)

For each iteration, show:

- Current position and fitness
- Temperature
- Neighbor selected
- Whether move is accepted (with calculation if it's a worse move)

[7 marks]

(c) Based on your results from (a) and (b), explain:

- Why did Hill Climbing fail to find the global optimum?
- What characteristic of Simulated Annealing allows it to potentially escape local optima?
- In what temperature regime does Simulated Annealing behave most like Hill Climbing?

[4 marks]

Question 2: Local Search Convergence [10 marks]

Consider a local search algorithm on a Boolean space $\{0,1\}^n$ with the OneMax function: $\text{OneMax}(x) = \sum_{i=1}^n x_i$ (to be maximized).

- (a) If we use Hill Climbing with 1-bit flip neighborhood, prove that the algorithm will **always** reach the global optimum $x^* = (1, 1, \dots, 1)$ regardless of starting point. Show your reasoning clearly.

[5 marks]

(b) Now consider the TwoMax function:

$$\text{TwoMax}(x) = \max \left(\sum_{i=1}^n x_i, \sum_{i=1}^n (1 - x_i) \right)$$

Explain why Hill Climbing with 1-bit flip neighborhood may fail to reach the global optimum when starting from $x = (0, 1, 0, 1, \dots, 0, 1)$ (alternating bits). What is the basin of attraction issue here?

[5 marks]

Section 2: Runtime Analysis [40 marks]

Question 3: Tail Inequalities [15 marks]

Consider a random variable X representing the number of 1-bits in a uniformly random bit string of length $n = 500$.

- (a) Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.

[2 marks]

- (b) Use **Markov's inequality** to upper bound the probability $P(X \geq 350)$. Show all steps.

[3 marks]

- (c) Use **Chebyshev's inequality** to upper bound $P(X \geq 350)$. Show all steps.

[3 marks]

- (d) Use **Chernoff bound** (for the sum of independent Bernoulli trials) to upper bound $P(X \geq 350)$. Use the bound: $P(X \geq (1 + \delta)\mu) \leq e^{-\frac{\delta^2\mu}{3}}$ for $\delta \in [0, 1]$. Calculate δ first, then apply the bound. Show all steps.

[4 marks]

- (e) Compare the three bounds you obtained. Which one is tightest? Why does this make sense given the conditions and assumptions of each inequality?

[3 marks]

Question 4: (1+1) EA Runtime on LeadingOnes [15 marks]

Recall the LeadingOnes function: $LO(x) = \sum_{i=1}^n \prod_{j=1}^i x_j$

This function counts the number of leading ones before the first zero.

(a) For $n = 5$, calculate $LO(x)$ for the following bit strings:

- $x_1 = (1, 1, 1, 0, 1)$
- $x_2 = (1, 1, 0, 1, 1)$
- $x_3 = (1, 0, 1, 1, 1)$

[3 marks]

(b) Consider a bit string x with $LO(x) = k < n$. This means $x = (1, 1, \dots, 1, 0, x_{k+2}, \dots, x_n)$ where there are exactly k leading ones.

What is the probability that the (1+1) EA increases $LO(x)$ in one iteration?

(Hint: To improve, we must flip bit $k + 1$ from 0 to 1, and not flip any of the first k bits. The mutation rate is $1/n$.)

Show your calculation clearly.

[5 marks]

- (c) Using your result from (b), show that the expected time for the (1+1) EA to optimize LeadingOnes is:

$$\mathbb{E}[T] = \sum_{k=0}^{n-1} \mathbb{E}[T_k \rightarrow k+1] = O(n^2)$$

where $T_k \rightarrow k+1$ is the time to go from $\text{LO}(x) = k$ to $\text{LO}(x) = k+1$.

Show the summation and simplification steps.

[7 marks]

Question 5: Drift Analysis [10 marks]

Consider the (1+1) EA optimizing OneMax: $f(x) = \sum_{i=1}^n x_i$.

Let X_t denote the number of 0-bits at generation t . Our goal is to show that $\mathbb{E}[T] = O(n \log n)$ where T is the time until $X_T = 0$.

- (a) When $X_t = k$ (i.e., there are k zeros), what is the expected change $\mathbb{E}[X_t - X_{t+1} | X_t = k]$? This is called the "drift".

(Hint: Calculate the probability of improving exactly j bits for $j = 1, 2, \dots, k$, where improving means flipping a 0 to 1 without flipping any 1 to 0.)

For simplicity, you may use the approximation that the probability of flipping exactly one specific bit is $\frac{1}{n}(1 - \frac{1}{n})^{n-1} \approx \frac{1}{ne}$ and flipping multiple bits is less likely.

[6 marks]

- (b) Using the Additive Drift Theorem: If $\mathbb{E}[X_t - X_{t+1}|X_t] \geq \delta$ for all $X_t > 0$, then $\mathbb{E}[T] \leq \frac{X_0}{\delta}$.

However, our drift depends on k . Use the Multiplicative Drift Theorem instead: If $\mathbb{E}[X_t - X_{t+1}|X_t] \geq \delta \cdot X_t$ for all $X_t > 0$, then $\mathbb{E}[T] = O(\frac{\log X_0}{\delta})$.

Show that our drift satisfies this condition and conclude $\mathbb{E}[T] = O(n \log n)$.

[4 marks]

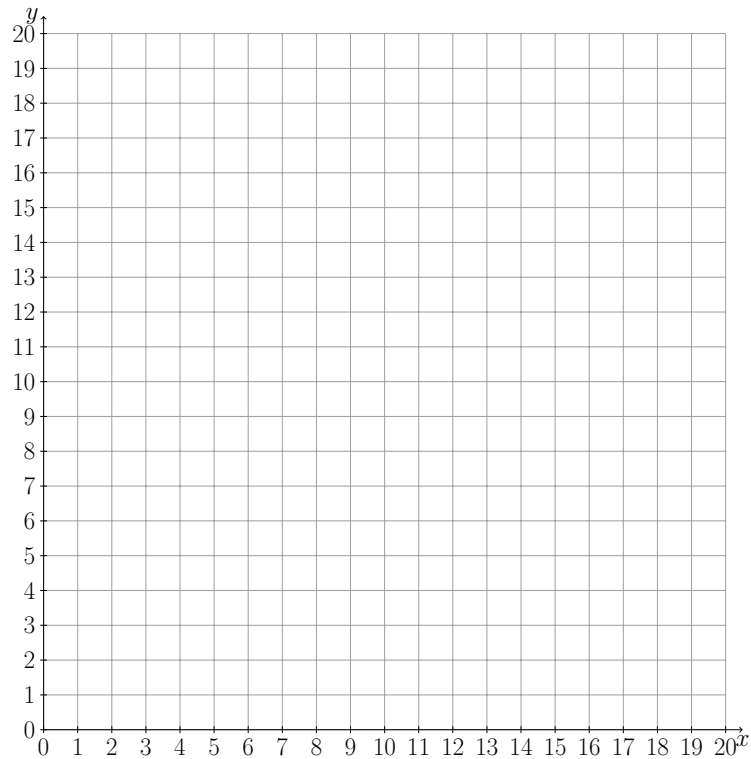
Section 3: Multi-Objective Optimization [45 marks]

Question 6: Pareto Dominance and Non-dominated Sorting [20 marks]

Consider a bi-objective maximization problem with the following 10 solutions and their objective values (f_1, f_2) :

Solution	f_1	f_2
A	8	12
B	10	10
C	6	15
D	12	8
E	10	10
F	9	11
G	7	13
H	11	9
I	8	10
J	13	7

- (a) Draw all 10 objective vectors in a 2D objective space (use graph paper or draw a clear grid). Label each point.



[4 marks]

- (b) Build a **dominance matrix**: For each pair of solutions, indicate if one dominates the other. Present your answer as a table where entry (i, j) shows whether solution i dominates solution j .

Use: “D” for dominates, “=” for equal (like B and E), “N” for neither dominates.

[6 marks]

- (c) Perform **non-dominated sorting** to partition the solutions into fronts $F_1, F_2, F_3, \dots$

For each solution, show:

- How many solutions dominate it (n_p = domination count)
- Which front it belongs to

Show your working step by step as the algorithm removes fronts iteratively.

[6 marks]

- (d) List all solutions in the first front F_1 (the Pareto front). For each solution in F_1 , explain why it is non-dominated.

[4 marks]

Question 7: NSGA-II Crowding Distance [15 marks]

Consider the first front F_1 you identified in Question 6.

- (a) Calculate the **crowding distance** for each solution in F_1 using the NSGA-II procedure:

- Solutions at the boundaries get infinite distance (∞)
- For each solution i in the middle:

$$d_i = \sum_{m=1}^M \frac{f_m^{i+1} - f_m^{i-1}}{f_m^{\max} - f_m^{\min}}$$

where $M = 2$ (two objectives), and $i + 1$, $i - 1$ are the neighbors in the sorted list for objective m .

Show all steps:

- Sort solutions by f_1 , assign boundary distances, calculate partial distances
- Sort solutions by f_2 , assign boundary distances, calculate partial distances
- Sum up the distances for each solution

[10 marks]

- (b) Suppose NSGA-II needs to select 5 solutions for the next generation, and all solutions from your F_1 must be compared. Using your crowding distances, which 5 would be selected? Explain the selection rule.

[3 marks]

- (c) What is the purpose of crowding distance in NSGA-II? How does it promote diversity?

[2 marks]

Question 8: Hypervolume Contribution [10 marks]

Using the Pareto front F_1 from Question 6, we will perform hypervolume-based selection.

Use reference point $r = (5, 5)$ (both objectives are to be maximized).

- (a) Draw the hypervolume dominated by F_1 with respect to reference point r . Shade the area clearly.

[3 marks]

- (b) Calculate the **exclusive hypervolume contribution** of each solution in F_1 . This is the hypervolume that would be lost if that solution were removed.

Show your calculation by:

- Computing total hypervolume with all solutions
- For each solution, computing hypervolume without it
- Subtracting to find the contribution

(Hint: For 2D hypervolume with points $(x_1, y_1), \dots, (x_k, y_k)$ sorted by x , the total hypervolume can be computed by summing rectangular regions.)

[5 marks]

- (c) If we need to remove 3 solutions iteratively (always removing the one with smallest hypervolume contribution), which solutions would be removed in which order? Show the contribution recalculation after each removal.

[2 marks]

Section 4: Evolutionary Submodular Optimization [40 marks]

Question 9: Submodular Function Properties [15 marks]

- (a) Define precisely what it means for a set function $f : 2^V \rightarrow \mathbb{R}$ to be **submodular**. Give both the equivalent definitions:
- The diminishing returns property
 - The inequality formulation

[4 marks]

(b) Consider the set $V = \{a, b, c, d\}$ and the function f with values:

Set S	$f(S)$
$\emptyset$	0
$\{a\}$	4
$\{b\}$	3
$\{c\}$	5
$\{d\}$	2
$\{a, b\}$	6
$\{a, c\}$	8
$\{a, d\}$	5
$\{b, c\}$	7
$\{b, d\}$	4
$\{c, d\}$	6
$\{a, b, c\}$	10
$\{a, b, d\}$	7
$\{a, c, d\}$	9
$\{b, c, d\}$	8
$\{a, b, c, d\}$	11

Verify whether f is submodular by checking the diminishing returns property for:

- Adding element d to $\{a\}$ vs. adding d to $\{a, b\}$
- Adding element c to $\{b\}$ vs. adding c to $\{b, d\}$

Show your calculations. Is f submodular based on these checks?

[6 marks]

- (c) The function f is also **monotone** (i.e., $f(A) \leq f(B)$ whenever $A \subseteq B$).

State the greedy algorithm for maximizing a monotone submodular function under a cardinality constraint $|S| \leq k$. Then explain why this greedy algorithm achieves a $(1 - 1/e)$ -approximation.

(You don't need to prove it rigorously, but explain the intuition behind the approximation guarantee.)

[5 marks]

Question 10: Maximum Coverage - Graph Analysis [15 marks]

Consider the graph $G = (V, E)$ where:

$$V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

and

$$E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_4\}, \{v_3, v_4\}, \{v_4, v_5\}, \{v_5, v_6\}\}$$

- (a) Draw this graph clearly.

[2 marks]

- (b) For the **maximum coverage problem**, we want to select a subset of nodes such that the number of covered nodes is maximized. A node is covered if it is selected or if it has an edge to a selected node.

Consider the bi-objective formulation where we:

- Maximize $f_1(S)$ = number of covered nodes
- Minimize $f_2(S)$ = number of selected nodes

For each of the following solutions (bit strings where $x_i = 1$ means node v_i is selected), calculate (f_1, f_2) :

Solution	Selected nodes	(f_1, f_2)
000000	$\emptyset$	
100000	$\{v_1\}$	
010000	$\{v_2\}$	
001000	$\{v_3\}$	
000100	$\{v_4\}$	
110000	$\{v_1, v_2\}$	
100100	$\{v_1, v_4\}$	
010100	$\{v_2, v_4\}$	

Fill in the table.

[5 marks]

(c) Identify ALL Pareto optimal solutions for this graph (there may be more than those listed above). For each Pareto optimal solution, give:

- The bit string representation
- The selected nodes
- The objective vector (f_1, f_2)

Explain why each is Pareto optimal.

[5 marks]

(d) Draw the Pareto front in objective space (i.e., plot all Pareto optimal objective vectors with f_1 on the x-axis and f_2 on the y-axis).

Note: Since we're minimizing f_2 , the Pareto front should show the trade-off between coverage (high f_1) and cost (low f_2).

[3 marks]

Question 11: GSEMO Runtime for Maximum Coverage [10 marks]

Consider the path graph $G = (V, E)$ where $V = \{v_1, \dots, v_n\}$ and $E = \bigcup_{i=1}^{n-1} \{\{v_i, v_{i+1}\}\}$ (a simple path).

For the bi-objective maximum coverage formulation, consider using GSEMO (Global Simple Evolutionary Multi-objective Optimizer).

- (a) What is the size of the Pareto front for this path graph? (I.e., how many Pareto optimal solutions are there?) Explain your reasoning.

[4 marks]

- (b) Starting from the empty solution (all nodes not selected), what is the probability that GSEMO finds a solution covering all nodes in a single mutation? (Use mutation rate $1/n$ per bit.)

[3 marks]

- (c) Using fitness level arguments (or partition-based analysis), show that GSEMO finds a solution covering all nodes in expected time $O(n^2)$.

(Hint: Consider the levels based on number of covered nodes, and analyze the probability of improving the coverage.)

[3 marks]

Section 5: Integrated Understanding [20 marks]

Question 12: Multi-objective LOTZ Problem [12 marks]

Recall the multi-objective problem LOTZ where:

$$\text{LOTZ}(x) = (\text{LO}(x), \text{TZ}(x))$$

where $\text{LO}(x) = \sum_{i=1}^n \prod_{j=1}^i x_j$ (leading ones) and $\text{TZ}(x) = \sum_{i=1}^n \prod_{j=n-i+1}^n (1-x_j)$ (trailing zeros).

Both objectives should be maximized.

(a) For $n = 6$, calculate $\text{LOTZ}(x)$ for:

- $x_1 = (1, 1, 1, 0, 0, 0)$
- $x_2 = (1, 1, 0, 0, 0, 0)$
- $x_3 = (0, 0, 0, 0, 0, 0)$
- $x_4 = (1, 1, 1, 1, 1, 1)$

[4 marks]

(b) Give ALL Pareto optimal solutions for $n = 6$ in terms of their form. (Hint: Think about the structure of solutions that maximize one or both objectives simultaneously.)

[4 marks]

(c) How many Pareto optimal objective vectors are there for general n ? Explain.

[4 marks]

Question 13: Conceptual Understanding [8 marks]

Answer the following short questions:

- (a) In NSGA-II, why do we use crowding distance instead of simply using random selection to choose among solutions in the same front?

[2 marks]

- (b) Why is the $(1 + 1)$ EA expected to have $O(n^2)$ runtime on LeadingOnes but $O(n \log n)$ on OneMax? What structural difference between these functions causes this?

[3 marks]

- (c) In submodular optimization, why does the greedy algorithm provide a $(1 - 1/e) \approx 0.632$ approximation rather than something like 0.5 or 0.9? What mathematical property drives this specific value?

[3 marks]

— END OF EXAMINATION —